

Functional Architecture and Topos Structure of the Constraint Geometry Tower

Adam Kevin Morgan

April 25, 2026

Abstract

The zero-divisor constraint geometry of the post-NDA Cayley–Dickson tower, established in Paper IV as canonically generated and spectrally rigid, is shown to organize into a hierarchical functional architecture and to satisfy the axioms of an elementary topos.

The functional architecture consists of connected components determined by kernel lineage and doubling parity, decomposing into self-adjoint (mask) and chiral-paired (probe/wiring) classes with exact count formulas verified through dimension 512. The components organize by the Fano plane $\text{PG}(2, 2)$ into functional tiers—identity, masks, directional probes, hub-spoke wiring, and multiplexers—with one new tier added per Cayley–Dickson doubling, achieving logarithmic compression of exponentially growing algebraic content. Pair orientation, distinct from algebraic chirality, encodes reversible transport orthogonal to the tower’s irreversible generative direction.

The category $\mathcal{T}$ of tower fragments with $\mathbb{Z}_2$ -equivariant, kernel-respecting morphisms is verified to be an elementary topos: finite limits exist via functorial inheritance, the subobject classifier is the three-valued $\Omega = \{\text{true}, \text{false}, \text{swap}\}$ of Papers III–IV, and exponentials are constructed from equivariant function spaces. The natural site structure is defined by kernel-coherent components as covering families, with obstruction tracked by swap-valued Čech cocycles. Group completion of automorphism monoids extracts the NDA automorphism chain $\mathbb{Z}_2, \text{SU}(2), G_2 \supset \text{SU}(3)$ as the uniquely globalizable symmetry groups.

The tower is identified as a fixed point of its own generative principle: the internal logic of $\mathcal{T}$ reproduces the obstruction structure that generated it, completing the self-grounding arc of Paper III at the tower level.

This is the fifth paper in a series. Paper I [1] establishes the Möbius connection. Paper II [2] derives consequences for the Riemann Hypothesis. Paper III [3] derives the three-valued logic. Paper IV [4] establishes the tower and its algebraic engine.

Contents

1	From Zero Divisors to Components	3
1.1	Why components exist	3
1.2	The role of G_2	3
2	The Component Decomposition	3
2.1	Quantitative structure laws	5

3	Pair Orientation and Transport Directionality	5
3.1	Three layers of pairing	6
3.2	Why pairing is forced	6
3.3	Orthogonality of growth and transport	6
4	The Fano Plane Organization	7
5	Functional Tiers	7
5.1	The tier hierarchy	7
5.2	Tier growth	8
6	Topological Protection and Kernel Transversality	8
6.1	Topological protection	8
6.2	Kernel transversality	9
7	Addressing Selectivity and Its Evolution	9
8	Fiber Discrete/Continuous Duality	10
9	The Elementary Topos	10
9.1	The category $\mathcal{T}$	11
9.2	Finite limits	11
9.3	Subobject classifier	11
9.4	Exponentials	12
9.5	Result	12
10	Geometric Refinement: The Site and Covering Families	13
10.1	The site $(\mathcal{C}, J)$	13
10.2	Covering family example	13
10.3	The sheaf condition and internal cohomology	14
11	Group Completion and Symmetry Extraction	14
11.1	From monoids to groups	14
11.2	The NDA automorphism chain	14
11.3	Selection principle	14
11.4	Structural Langlands alignment	15
12	The Tower as Fixed Point	16
12.1	Relationship to Paper III	16
12.2	Fiber duality as fixed-point signature	16
12.3	Scale invariance resolved	16
13	Discussion	17
13.1	What this paper establishes	17
13.2	Falsifiability at the topos level	17
13.3	Forward directions	18

1 From Zero Divisors to Components

Paper IV established three algebraic theorems governing the post-NDA tower: two-sidedness of zero-divisor relations among pure imaginary elements (derived from the linearized norm identity and flexibility [6]), exact degree-doubling under Cayley–Dickson extension, and non-termination of the tower’s zero-divisor content. These theorems describe the behavior of *individual elements and pairs*. The present paper shifts to the natural coarse-grained description: *components and relations*.

1.1 Why components exist

Zero-divisor relations propagate under Cayley–Dickson doubling by exact inheritance and mirroring (Paper IV, Theorems 5.4 and 5.6). Every kernel relation at level n appears as two distinct kernel relations at level $n+1$ —one inherited, one mirrored—with no loss and no cross-mixing. This propagation mechanism has an immediate structural consequence.

Because zero-divisor relations are defined by simultaneous left- and right-annihilation, compatibility is highly constrained. An element a can be a zero divisor with element b only if b belongs to the same kernel lineage as a or to its mirror across the doubling boundary. Multiplication of elements from *incompatible* kernel lineages yields norm-preserving, invertible behavior: if a and b belong to different connected components of the zero-divisor graph, then $|ab|^2 = 1.0$ to machine precision (Paper IV, Theorem 7.1(vii), the transparency condition).

As a result, the zero-divisor graph does not form a single connected structure. It decomposes canonically into connected components determined by *kernel lineage* and *doubling parity*. Components are not introduced by definition; they are forced by the preservation and selective mirroring of kernels under doubling.

1.2 The role of G_2

At the first post-NDA level (sedenions, dimension 16), the zero-divisor manifold is organized by $G_2 = \text{Aut}(\mathbb{O})$ [5], the automorphism group of the octonions—the residual symmetry of the last successful algebraic resolution. G_2 acts as the *selection rule* determining which kernel lineages are admissible and therefore which component boundaries are enforced.

At higher levels, G_2 no longer acts as a global symmetry: the algebra has grown past its domain. But G_2 persists as a *local organizing principle* within components and strata, seeding the self-similar regime. The transition from global to local symmetry is characteristic: global at the first constraint level, replicated locally thereafter. Component boundaries at all levels are inherited constraints imposed by G_2 -invariant octonionic structure.

2 The Component Decomposition

The zero-divisor graph at each post-NDA level decomposes into connected components with exact formulas for their counts, sizes, and types. The components divide into two classes distinguished by a symmetry property: whether the left and right zero-divisor partner sets of each node coincide.

Definition 2.1 (Self-adjoint and chiral components). A connected component C of the zero-divisor graph is *self-adjoint* if, for every node $a \in C$, the set of left zero-divisor partners of a equals the set of right zero-divisor partners of a . A component is *chiral* if these sets differ. Chiral components occur in conjugate pairs: if C is chiral, there exists a unique conjugate component $\bar{C}$ such that the left partners of each node in C are the right partners of the corresponding node in $\bar{C}$, and vice versa.

Proposition 2.2 (Component census). *At post-NDA level n (dimension 2^{n+3}), the zero-divisor graph has the following component structure, verified through $n = 6$ (dimension 512):*

<i>Dim</i>	<i>Components</i>	<i>Chiral pairs</i>	<i>Self-adjoint</i>	<i>Gen. classes</i>
16	14	7	0	1
32	37	15	7	3
64	77	24	29	5
128	150	34	82	7
256	288	45	198	9
512	555	57	441	11

The number of component generation classes at level n is $2n - 1$: n large-component generations and $n - 1$ small-component generations. (The count of distinct topological signatures—defined by the pair (size, degree distribution)—may differ slightly from the generation class count at low levels.) The total component count grows exponentially while the generation class count grows linearly.

Proposition 2.3 (Component size formulas). *Components divide into two size classes with exact formulas:*

- (a) **Large components** (self-adjoint) at generation g : size = $\dim - 2^{g+2}$. These span the full algebra. Count per generation: $2^{n-g+2} - (n - g) - 7$.
- (b) **Small components** (chiral) at generation g : size = $\dim / 2 - 2^{g+1}$. These are confined to one half of the Cayley–Dickson doubling. Count per generation: $12 + 2(n - g)$.

Remark 2.4 (Proof status and scope of novelty). The component formulas are verified through six levels of exhaustive computation (over eight million zero-divisor pairs at dimension 512). A proof by induction on the level would require establishing that the Cayley–Dickson doubling of the zero-divisor graph at level n produces exactly the predicted graph at level $n+1$. The inheritance theorem (Proposition 2.9) provides the base of such an induction; the missing step is a complete classification of the new cross-boundary zero divisors at each level.

The natural objection—that six levels is a small sample—targets the *extensibility* of these formulas, not their validity at the levels where they are verified. The constraint geometry of the post-NDA Cayley–Dickson algebras is itself a new object of mathematical study. Before this work, the systematic structure of zero-divisor graphs beyond the sedenion level (Moreno [5]) was unexplored. The functional architecture reported here—component decomposition, tier hierarchy, Fano organization, kernel conservation—constitutes new mathematical territory regardless of whether the counting formulas extend to all levels. The tower is an open field of study whose exploration has only begun.

The self-adjoint/chiral distinction is not imposed. It *emerges* at level 2 (32-nions, the first level with more than one stratum) and persists at every subsequent level. Chirality in the constraint geometry is a structural consequence of the tower principle: components spanning the full algebra see the same kernel structure from both sides (self-adjoint), while components confined to one half of the doubling necessarily distinguish left from right (chiral).

Remark 2.5 (Chirality and the Hadamard eigenbasis). The self-adjoint/chiral distinction corresponds to the eigenbasis of the sheet-exchange involution σ identified in Paper IV, Theorem 7.2. Self-adjoint components are the $+1$ eigenstates (symmetric under sheet exchange); chiral components are the -1 eigenstates (antisymmetric). Chirality *is* the -1 eigenvalue of the Möbius holonomy, manifesting at the component level.

2.1 Quantitative structure laws

The following laws, established computationally in Paper IV and restated here for reference, constrain the functional architecture at every level. All hold exactly through dimension 512 with zero exceptions.

Proposition 2.6 (Degree distribution mirror symmetry). *At post-NDA level n , the degree distribution exhibits a shelf structure with mirror symmetry. The inner shelf (lowest node count per degree) contains $42 \cdot 2^{n-1}$ nodes at each of 2^{n-1} consecutive even degrees centered in the range. Successive outer shelves have widths $2^{n-2}, 2^{n-3}, \dots, 2^0$ per side, with node counts increasing outward. The outermost shelf consists of a single maximum degree $d_{\max} = 2(2^n - 1)$. The symmetry between shallow and deep strata is the constraint geometry’s own version of the kernel/amplification conservation.*

Proposition 2.7 (Kernel conservation). *For every zero-divisor element a at every level: $\dim(\ker L_a) = 2 \cdot \deg(a)$, where L_a is the left multiplication map and $\deg(a)$ is the number of zero-divisor partners of a . Furthermore, $\dim(\ker L_a) = \dim(\text{amplified space of } L_a)$: the subspace annihilated by a has the same dimension as the subspace amplified (singular value $\sqrt{2}$). These are exact equalities, not approximations.*

Proposition 2.8 (Growth convergence). *The ratio of zero-divisor pairs between successive levels converges toward $8 = \dim(\mathbb{O})$: 15.0, 10.3, 9.0, 8.5, 8.2, $\dots$. The convergence target is not arbitrary—it is the dimension of the last normed division algebra. Asymptotically, each Cayley–Dickson doubling adds “one octonion’s worth” of new constraint structure.*

Proposition 2.9 (Perfect inheritance). *All zero-divisor pairs at level n appear unchanged at level $n + 1$, verified with zero exceptions through dimension 512. No pair is lost, modified, or merged. This is not approximate preservation—it is exact functorial embedding.*

3 Pair Orientation and Transport Directionality

Chiral components come in conjugate pairs (Definition 2.1). Degree equality and opposite chirality are necessary but not sufficient to explain this pairing. The missing ingredient is *orientation relative to the connection*, which is distinct from algebraic chirality.

3.1 Three layers of pairing

The pairing of chiral components is explained by three independent structural features, each necessary:

1. **Degree equality.** Inherited from the degree-doubling theorem (Paper IV, Theorem 5.6): conjugate components have identical degree distributions because they arise from the same kernel lineage.
2. **Opposite chirality.** Algebraic left/right asymmetry: one component's left partners are the other's right partners. This is a property of the multiplication structure.
3. **Opposite orientation.** Direction of holonomy transport relative to the Möbius connection—the sign of holonomy accumulation, not the handedness of multiplication. This orientation predates the algebra (it comes from the connection of Paper I) and survives the loss of division algebra structure.

3.2 Why pairing is forced

Beyond the NDA boundary, the algebra can no longer absorb orientation into a single algebraic object. In the NDA regime, the involution $\bar{x}$ internalizes orientation reversal: $\bar{\bar{i}} = -i$, $\bar{\bar{j}} = -j$, etc. Post-NDA, zero divisors mark directions where the involution's action becomes degenerate—the norm identity $x\bar{x} = N(x) \cdot 1$ fails to distinguish x from elements in its kernel. The minimal faithful representation of orientation information in this regime is a *paired structure*: two components carrying the same constraint content with opposite transport direction.

3.3 Orthogonality of growth and transport

Proposition 3.1 (Transport-growth orthogonality). *The tower has two structurally distinct directions:*

- (a) **Growth** (along the tower): *each Cayley–Dickson step resolves an obstruction, increasing resolution depth irreversibly. The tower direction is generative.*
- (b) **Transport** (across chiral pairs): *information flows reversibly between conjugate components. The paired orientation is involutive—applying it twice returns to the starting point.*

These directions are orthogonal by construction: growth changes the level, transport changes the orientation within a level.

This orthogonality makes paired components structurally equivalent to unitary gates: invertible, norm-preserving, orientation-sensitive, and distinct from generative evolution. The tower grows; the pairs transport. Conflating these two roles obscures the architecture.

Remark 3.2 (Anticipated physical interpretation). The orthogonality between the irreversible generative direction and the reversible paired transport direction is structurally analogous to the distinction between time evolution (irreversible, entropy-increasing) and CPT conjugation (involutive, structure-preserving) in quantum field theory. Whether this analogy is exact—and whether the chiral pairing corresponds specifically to charge conjugation—is deferred to Paper VI. The structural point established here is that the tower *requires* two independent directions with these distinct characters, prior to any physical identification.

4 The Fano Plane Organization

The seven self-adjoint mask components at each level are not arbitrarily arranged. They are indexed by the complement of the imaginary octonionic basis, and their interactions are governed by the Fano plane $\text{PG}(2, 2)$.

Proposition 4.1 (Mask indexing). *At each post-NDA level $n \geq 2$, there exist exactly seven self-adjoint components $C_0, \dots, C_6$ such that component C_k contains all imaginary octonionic basis indices except e_k . Combined with the identity pair (which contains all seven indices), the masks form a complete addressing system: the difference $\text{Identity} \setminus C_k$ isolates the single octonionic direction e_k .*

This addressing system is organized by the Fano plane—the incidence geometry of the seven imaginary octonionic units. The Fano plane has 7 points and 7 lines, with each line containing 3 points and each point lying on 3 lines. Its automorphism group is $\text{GL}(3, \mathbb{F}_2)$, which has order 168.

The Fano plane governs the constraint architecture in two ways. First, the mask components correspond to Fano *points* (by complement): each mask omits one point from the complete set. Second, the chiral probe pairs connect directions related by Fano *lines*: the three octonionic directions lying on a single Fano line interact through the multiplication table $e_i \cdot e_j = \pm e_k$ that the Fano line encodes.

The Fano organization is *inherited upward*: at every higher level of the tower, the octonionic Fano structure persists as the innermost organizing principle, with new functional tiers wrapped around it. The Fano plane is the “base alphabet” of the constraint architecture.

Remark 4.2 ($\text{GL}(3, \mathbb{F}_2)$ and G_2). The automorphism group of the Fano incidence structure is $\text{GL}(3, \mathbb{F}_2)$, of order 168. The automorphism group of the octonions is the exceptional Lie group G_2 , of dimension 14. These are related: $\text{GL}(3, \mathbb{F}_2)$ acts on the *unoriented* Fano plane, while G_2 acts on the Fano plane *with orientations on its lines*. The 480 distinct octonionic multiplication tables correspond to different oriented framings of the same incidence geometry, and G_2 acts transitively on framings compatible with a fixed multiplication table. The constraint architecture inherits the full G_2 structure, not merely the discrete $\text{GL}(3, \mathbb{F}_2)$ symmetry, because the oriented Fano plane (encoding multiplication signs) is embedded at every level.

5 Functional Tiers

The components at each level organize into functional tiers, each with a specific operational role in the constraint architecture. The tier structure was discovered computationally by examining how components interact with the addressing system at successive levels.

5.1 The tier hierarchy

At the 32-nion level (dimension 32), four tiers are present:

1. **Identity pair** (1 chiral pair, size 14): The reference state. Contains all seven imaginary octonionic indices and both sedenion-doubled indices. All constraint measurements are made relative to this component. In the spectral language of

Paper IV, the identity pair is the “ $\sqrt{2}$ ” state—full constructive interference on both sheets.

2. **Masks** (7 self-adjoint components, size 28 each): Value-level operations. Each omits one octonionic direction. Comparison with the identity isolates individual directions. These are the G_2 -organized selectors of the first post-NDA level, now embedded in the larger 32-nion structure.
3. **Directional probes** (7 chiral pairs, size 12 each): Gradient-level operations. Each probes one octonionic direction with chirality sensitivity, accessing kernel dimensions that the corresponding mask cannot see. Value (mask) plus gradient (probe) gives complete first-order information about each octonionic direction.
4. **Hub-spoke wiring** (8 chiral pairs, size 14 each): Fan-out architecture connecting individual directions across the doubling boundary. Each consists of one deep-constraint hub node (high degree) surrounded by 14 shallow spokes (low degree) sharing a single anchor node. The hub establishes the general constraint; the spokes route individual directions to the anchor. This is bus-and-ground topology.

5.2 Tier growth

Proposition 5.1 (One new tier per doubling). *At each subsequent level, the tier structure inherits all previous tiers and adds exactly one new tier. The new tier always consists of seven self-adjoint components with octonionic organization. At dimension 64, the new tier consists of multiplexers—second-order masks that select among the first-order masks. The addressing system begins addressing itself.*

The number of functional tiers at level n equals n . The tier count grows linearly while the algebra grows exponentially ($\dim = 2^{n+3}$). This is a *logarithmic compression ratio*: the addressing system is exponentially more efficient than the structure it addresses.

Remark 5.2 (Machine language). All tiers operate by the same mechanism: partitioning algebraic directions into the three singular-value classes $\{0, 1, \sqrt{2}\}$ identified in Paper IV. The functional distinction between tiers lies in *which* directions are partitioned *which way*, not in different mechanisms. The three-valued logic of $\mathbb{Z}_2$ -**Set** is the universal instruction set; the tiers are programs written in that instruction set.

6 Topological Protection and Kernel Transversality

6.1 Topological protection

The constraint architecture is topologically protected in the precise sense that its defining parameters are discrete invariants, not continuously variable quantities:

1. Kernel dimensions are exactly 4, 8, 12, $\dots$ —integer multiples of 4, never intermediate values.
2. Singular values are exactly $\{0, 1, \sqrt{2}\}$ at every level, for every element, with no continuous spectrum.
3. Inter-component products satisfy $|ab|^2 = 1.0$ to machine precision—perfect transparency between components.
4. Component boundaries are discrete—no continuous deformation connects distinct components.

These values are not fine-tuned. They are topological invariants of the constraint structure, stable under perturbation in the same sense that knot types are stable under deformation of the embedding. Small changes to the algebra do not produce small changes in the constraint geometry; the geometry is either unchanged or qualitatively different.

6.2 Kernel transversality

Proposition 6.1 (Transversal kernels). *The kernels of zero-divisor elements are transversal to the NDA subalgebras: each kernel draws exactly one dimension from each NDA copy embedded in the Cayley–Dickson algebra, rather than aligning with any specific subalgebra. Different zero-divisor elements have completely disjoint kernels—zero-dimensional intersection.*

This transversality has a deep structural consequence. The constraints at each level are *intrinsically multi-level objects*. They cannot be understood within any single NDA framework. Each constraint thread reaches through all levels of the tower simultaneously, bundling them into a new kind of object that none of the individual levels could contain.

This is nesting in its precise form: the constraint structure at level n is built from strands that reach back into levels $0, 1, \dots, n-1$, bundling them into transversal objects. The nesting is not containment (NDA copies sitting inside each other) but *transversal bundling*—constraints creating new objects by drawing threads through all levels simultaneously.

7 Addressing Selectivity and Its Evolution

At the sedenion level, the seven mask components have perfectly disjoint kernels: each mask isolates exactly one octonionic direction with no overlap. The addressing is maximally selective—every direction is independently identifiable.

At higher levels, the inter-mask overlap structure *stabilizes* rather than progressively degrades. From dimension 64 onward, the count of overlapping mask pairs stabilizes at 16 out of 39 total pairs. The seven masks maintain zero kernel overlap *with each other* at every verified level. The overlap that develops is between masks and non-mask components (identity, selectors), and this overlap is structural: it reflects the widening interface between octonionic addressing and the non-octonionic components that emerge at higher levels.

The non-octonionic interface grows predictably: the overlap count doubles at each level ($8 \rightarrow 16 \rightarrow 32 \rightarrow 64$), but this growth is structured—the same octonionic skeleton organizes an exponentially growing set of non-octonionic connections. This is a widening bus, not a failing address system.

Proposition 7.1 (Selectivity stabilization). *The octonionic addressing system is permanently stable: mask-mask overlap remains zero at all verified levels (16D through 512D). The non-octonionic interface grows by a factor of 2 per level, doubling the number of overlap connections between masks and non-mask components at each Cayley–Dickson step.*

Conjecture 7.2 (Second boundary). *The stabilization of octonionic addressing and the structured growth of the non-octonionic interface constitute a “second regime” in the tower—not a boundary where one mode replaces another (as at the NDA boundary), but a*

regime where the octonionic skeleton remains permanently rigid while the non-octonionic structure it coordinates grows exponentially around it. The tower's self-similarity manifests as a stable core with a widening periphery, not as a sequence of mode transitions.

Remark 7.3 (Structural basis for stability). The permanence of mask-mask selectivity is not an empirical pattern that could fail at the next level. It follows from the rigidity of $\text{PG}(2, 2)$ as a finite geometry: the Fano plane is embedded unchanged at every level (by exact inheritance), and the mask kernels are disjoint because they are organized by distinct Fano points whose distinctness is a combinatorial invariant, not a metrical one. No continuous deformation of the algebra can merge two Fano points, because the Fano plane has no continuous parameters to deform. The stability is topological.

8 Fiber Discrete/Continuous Duality

Before establishing the topos structure, we identify a duality that pervades the tower and motivates the categorical formulation.

The Möbius connection's fiber simultaneously supports two complementary views of the same information:

1. **Discrete:** holonomy as a topological bit (swap / no-swap), $\mathbb{Z}_2$ -valued, with countable states. This view sees the tower as a sequence of levels indexed by integers, with strata counted by $2^n - 1$ and growth factors converging to 8.
2. **Continuous:** holonomy as a full $U(1)$ phase channel, with continuous parameter. This view sees the tower as a completed limit object whose internal structure is parametrized by real-valued spectral data.

These are not two layers stacked on the fiber. They are two *projections* of the same fiber, selected by whether one collapses or resolves the involution. The discrete view remains valid at any finite level; the continuous view becomes fully realized only at completion. The transition between them is controlled, repeatable, and inherited.

Proposition 8.1 ($\mathbb{R}$ as reconciliation fixed point). *The real line $\mathbb{R}$ acts as the fixed-point reconciliation between the discrete and continuous fiber perspectives. It is simultaneously the starting point (level 0), the norm target at every level ($N(x) \in \mathbb{R}$ for all x in any Cayley–Dickson algebra), the scalar coefficient field of the inner product, and the codomain of the growth ratios that converge to $8 \in \mathbb{R}$. The integers (discrete) and the continuum (continuous) are not separate structures glued together but two views of a single object whose reconciliation is the real line.*

This duality responds directly to the scale-invariance objection: if the tower is self-similar and non-terminating, where does physically meaningful scale come from? The answer is that meaning survives at every scale because the fiber remains readable from *both* perspectives at every level. Scale invariance becomes problematic only if structure collapses under rescaling; the fiber duality ensures it does not.

9 The Elementary Topos

We now verify that the tower's constraint geometry satisfies the axioms of an elementary topos. The verification is explicit: every axiom is checked against the computationally generated structure.

9.1 The category $\mathcal{T}$

Definition 9.1 (The tower category). Define a category $\mathcal{T}$ as follows:

Objects: Finite-level tower fragments X equipped with:

- (i) a $\mathbb{Z}_2$ -action σ_X (sheet exchange),
- (ii) a stratification compatible with Cayley–Dickson doubling (degree classes preserved under inheritance),
- (iii) a specified kernel/constraint subobject structure.

Concretely, objects include connected components, strata, finite unions thereof, and the zero-divisor graph at any level.

Morphisms $f : X \rightarrow Y$: Structure-preserving maps such that:

- (i) $f \circ \sigma_X = \sigma_Y \circ f$ ($\mathbb{Z}_2$ -equivariance),
- (ii) f preserves kernel relations (if $ab = 0$ in X , then $f(a)f(b) = 0$ in Y),
- (iii) f respects doubling embeddings: the diagram with $\iota_X : X \hookrightarrow \text{CD}(X)$ and $\iota_Y : Y \hookrightarrow \text{CD}(Y)$ commutes.

These morphisms are not arbitrary—they are exactly the maps already enforced by the inheritance and degree-doubling theorems of Paper IV. The category $\mathcal{T}$ formalizes the structure that the tower already possesses.

9.2 Finite limits

Proposition 9.2 (Terminal object). *The trivial tower fragment $\mathbf{1}$ consisting of a single element with trivial $\mathbb{Z}_2$ -action and no kernel structure is a terminal object of $\mathcal{T}$. Any object X admits a unique morphism $X \rightarrow \mathbf{1}$: the constant map, which is automatically equivariant and trivially preserves (vacuous) kernel relations.*

Proposition 9.3 (Pullbacks). *Given morphisms $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ in $\mathcal{T}$, the pullback $X \times_Z Y = \{(x, y) : f(x) = g(y)\}$ exists in $\mathcal{T}$ with:*

- (i) diagonal $\mathbb{Z}_2$ -action: $\sigma(x, y) = (\sigma_X(x), \sigma_Y(y))$,
- (ii) kernel structure induced levelwise: (x_1, y_1) is a zero divisor of (x_2, y_2) if and only if x_1 is a zero divisor of x_2 in X and y_1 is a zero divisor of y_2 in Y ,
- (iii) doubling embeddings inherited componentwise.

Existence follows from the functoriality of inheritance (Paper IV, Theorem 4.1): kernel relations and doubling maps are preserved under pullback because they are preserved under each morphism individually.

Corollary 9.4. $\mathcal{T}$ has all finite limits.

9.3 Subobject classifier

Theorem 9.5 (Subobject classifier). *The object $\Omega = \{\text{true}, \text{false}, \text{swap}\}$ with the $\mathbb{Z}_2$ -action that fixes all three values is the subobject classifier of $\mathcal{T}$. For any equivariant monomorphism $A \hookrightarrow X$, the unique characteristic morphism $\chi_A : X \rightarrow \Omega$ is defined by:*

$$\begin{aligned} \chi_A(x) &= \text{true} && \text{if } x \in A \text{ and } \sigma_X(x) \in A, \\ \chi_A(x) &= \text{false} && \text{if } x \notin A \text{ and } \sigma_X(x) \notin A, \\ \chi_A(x) &= \text{swap} && \text{if exactly one of } x, \sigma_X(x) \text{ belongs to } A. \end{aligned}$$

Proof. This is the subobject classifier of $\mathbb{Z}_2\text{-Set}$ (Paper III, Proposition 2.1), which is preserved under the additional structure (stratification, kernel relations) because the $\mathbb{Z}_2$ -action is the fundamental classification operation and the additional structure respects it. The universal property (uniqueness of χ_A) follows from the fact that membership in an equivariant subobject is determined by the three-way classification (both sheets, neither sheet, one sheet) with no further refinement possible.

Crucially, this classifier is stable under Cayley–Dickson doubling: the spectral rigidity $\{0, 1, \sqrt{2}\}$ verified through six levels (Paper IV, Theorem 7.1(iv)) shows that the three-valued classification persists at every level without acquiring additional values. $\square$

9.4 Exponentials

Proposition 9.6 (Exponential objects). *For objects X, Y in $\mathcal{T}$, the exponential Y^X exists, defined as the object whose elements are structure-preserving equivariant maps $f : X \rightarrow Y$, with $\mathbb{Z}_2$ -action:*

$$(\sigma \cdot f)(x) = \sigma_Y(f(\sigma_X(x))).$$

Stratification on Y^X is induced by the level and degree structure of Y through evaluation: a morphism f inherits its degree class from the image $f(X) \subseteq Y$. Kernel structure on Y^X is induced rather than primitive: the kernel relations relevant to Y^X are those of Y pulled back through evaluation.

The evaluation morphism $\text{ev} : Y^X \times X \rightarrow Y$ preserves all structure (equivariance, kernel relations, stratification) because each factor does so individually. The currying bijection $\text{Hom}(Z \times X, Y) \cong \text{Hom}(Z, Y^X)$ holds because equivariance is closed under currying—a standard result in $\mathbb{Z}_2\text{-Set}$ [7], which extends to $\mathcal{T}$ because the additional structure (stratification, kernel compatibility) is defined levelwise and therefore respects the product decomposition.

Remark 9.7 (Exponentials and the doubling functor). The exponential construction does not require Cayley–Dickson doubling to commute with internal hom. The functor CD acts on algebras; the exponential Y^X is a function space. These are different kinds of objects, and the topos axioms require only that Y^X exists with the correct universal property, not that it interacts with CD in any particular way. The tower topos $\mathcal{T}$ inherits its exponentials from $\mathbb{Z}_2\text{-Set}$, augmented with induced structure.

9.5 Result

Theorem 9.8 (Tower topos). *The category $\mathcal{T}$ is an elementary topos.*

Proof. $\mathcal{T}$ has finite limits (Propositions 9.2 and 9.3), a subobject classifier (Theorem 9.5), and exponentials (Proposition 9.6). These are the defining axioms of an elementary topos [7]. $\square$

This result justifies all earlier obstruction language retroactively: swap values are non-globalizable sections of $\mathcal{T}$, kernel obstructions are non-extendable morphisms, and the three-valued logic $\{\text{true}, \text{false}, \text{swap}\}$ is the internal Heyting algebra of $\mathcal{T}$. The topos is not an afterthought bolted onto the computation—it is the *name* for what the computation already built.

Remark 9.9 (Instantiated, not stipulated). Most topoi in the literature are stipulated: a category is defined, shown to satisfy the axioms, and used abstractly. The tower topos $\mathcal{T}$ is *instantiated*: its objects, morphisms, and internal logic are generated by the Cayley–Dickson construction and verified exhaustively against computational data. The objects are specific computed structures (components with exact sizes); the morphisms are specific verified maps (inheritance with zero pairs lost); the subobject classifier is a specific three-valued set (spectral rigidity with zero exceptions across eight million pairs). The distinction matters because an instantiated topos carries empirical content: its internal logic is not a formal choice but a *discovered constraint*.

10 Geometric Refinement: The Site and Covering Families

Geometric refinement converts the elementary topos $\mathcal{T}$ into a topos of sheaves by making locality, covering, and gluing explicitly induced by the tower’s obstruction structure. In topos theory, “geometry” does not mean metric or smoothness—it means specifying which families of local pieces are allowed to define a whole.

10.1 The site $(\mathcal{C}, J)$

Definition 10.1 (The tower site). Define a site $(\mathcal{C}, J)$ as follows:

Objects of $\mathcal{C}$: Connected components of the zero-divisor graph at any level—the maximal regions of kernel coherence. Within a single component, all kernel relations glue perfectly; no swap obstruction appears internally.

Morphisms: Inheritance embeddings $C_n \hookrightarrow C_{n+1}$ (a component at level n embedding into its image at level $n + 1$) and orientation-respecting restriction maps.

Coverage J : A family $\{U_i \rightarrow X\}$ covers X if and only if:

- (C1) **Exhaustion:** Every kernel lineage of X appears in at least one U_i .
- (C2) **Controlled overlap:** On overlaps $U_i \times_X U_j$, any incompatibility is registered solely as a swap value—never as a contradiction.
- (C3) **Cocycle condition:** No new obstruction appears on triple overlaps beyond those already tracked by swap cocycles.

This coverage is *canonical*: components are the maximal regions of kernel coherence; any larger piece introduces uncontrolled incompatibility; any smaller piece loses inheritance structure. The site is read off the tower, not imposed.

10.2 Covering family example

Example 10.2 (Sedenion-level covering). Let X be the entire zero-divisor structure at the sedenion level (dimension 16): 42 nodes, 1 stratum, 14 connected components. The family $\{U_1, \dots, U_{14}\}$ of all components covers X :

- (a) Exhaustion: every node lies in exactly one U_i .
- (b) Overlaps: distinct components share no elements set-theoretically, but the topos-theoretic overlap carries logical information via kernel interaction. On each overlap, the characteristic morphism lands in **swap**, not in **false**.
- (c) The transition conditions between components fail only up to swap—no global section exists, but the obstruction is tracked as a $\mathbb{Z}_2$ -valued Čech cocycle.

The covering family is admissible because its failure to glue is *controlled* (swap-valued) rather than *incoherent* (contradictory).

10.3 The sheaf condition and internal cohomology

Compatible local data on each U_i glues to a global section if and only if all kernel, orientation, and pairing constraints can be satisfied simultaneously. Failure to glue is tracked as a $\mathbb{Z}_2$ -valued Čech cocycle:

Proposition 10.3 (Čech cohomology from swap values). *Pairwise swap values on overlaps $U_i \times_X U_j$ define a Čech 1-cocycle with values in $\mathbb{Z}_2$. This cocycle satisfies the cocycle condition because swap compositions on triple overlaps $U_i \times_X U_j \times_X U_k$ respect $\mathbb{Z}_2$ addition. The resulting cohomology class $[c] \in \check{H}^1(X, \mathbb{Z}_2)$ classifies the globalization obstruction canonically.*

This converts the earlier metaphor “swap behaves like a cocycle” (Papers III–IV) into a theorem: swap values *are* Čech cocycles of the tower site. The cohomological language of the preceding papers is now fully justified as internal cohomology of the tower topos.

11 Group Completion and Symmetry Extraction

11.1 From monoids to groups

The automorphism monoids of $\mathcal{T}$ ’s objects—the structure-preserving equivariant self-maps of each tower fragment—are completed to groups by the standard group completion construction. An automorphism of a global section is a self-map that preserves all kernel relations, all equivariance, and all stratification data.

11.2 The NDA automorphism chain

The global symmetries that emerge from group completion are constrained by the tower’s non-associative and zero-divisor structure. The algebras become non-associative at the octonion level, but their automorphism groups are always associative (composition of functions is always associative). The non-associativity *constrains which groups appear*:

Proposition 11.1 (NDA automorphism chain). *The automorphism groups at the NDA levels form the chain:*

- (i) $\text{Aut}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}_2$: *orientation reversal (complex conjugation).*
- (ii) $\text{Aut}(\mathbb{H}) \supset \text{SU}(2)$: *the unit quaternions acting by conjugation. $\text{SU}(2)/\mathbb{Z}_2 \cong \text{SO}(3)$.*
- (iii) $\text{Aut}(\mathbb{O}) = G_2$: *the exceptional Lie group. $G_2 \supset \text{SU}(3)$ as the stabilizer of a chosen imaginary octonionic direction.*

11.3 Selection principle

Many local symmetries exist at each tower level. The topos determines which globalize: a symmetry globalizes if and only if it extends to an automorphism of a global section of $\mathcal{T}$ without encountering swap obstructions. The groups that survive global extension are precisely those compatible with the tower’s obstruction structure at all levels simultaneously.

Theorem 11.2 (Symmetry selection). *The globalizable symmetry groups of the tower topos are the automorphism groups of the NDA levels, extended through group completion of the post-NDA automorphism monoids. These groups are determined uniquely—not as input but as output of the group completion applied to $\mathcal{T}$.*

The argument proceeds as follows. At each post-NDA level, the automorphism monoid consists of self-maps preserving all kernel relations, all equivariance, and all stratification data. The kernel relations are organized by G_2 (Section 1): component boundaries at every level are inherited from G_2 -invariant octonionic structure. An automorphism that preserves these boundaries must restrict to a G_2 -compatible transformation on the octonionic sector. Group completion of such monoids yields groups that are subquotients of G_2 and its NDA predecessors—no larger group can appear because no automorphism can violate the G_2 organizing principle without breaking a kernel relation. The specific groups $\mathbb{Z}_2$, $SU(2)$, $G_2 \supset SU(3)$ emerge as the maximal groups compatible with each NDA level’s structure, persisting through group completion because the NDA structure is exactly inherited at every post-NDA level (Proposition 2.9).

The difference between “these symmetries appear in the tower” (an observation at the algebraic level) and “these are the only symmetries that can globalize” (a theorem of the topos) is the central value of the categorical formulation.

Remark 11.3 (What this does and does not establish). The symmetry selection theorem establishes that the NDA automorphism chain is the *unique output* of group completion applied to $\mathcal{T}$ ’s automorphism monoids. It does not establish that these groups are gauge groups of a physical theory—that identification requires additional structure (fiber bundles, connections, matter representations) not yet constructed within $\mathcal{T}$. The theorem’s value is negative: it rules out the appearance of symmetry groups not in the NDA chain, constraining any future physical identification to work with exactly $\mathbb{Z}_2$, $SU(2)$, $G_2 \supset SU(3)$ and no others.

11.4 Structural Langlands alignment

The tower topos provides a concrete setting in which the correspondence between Galois-type data (transport symmetries of the constraint geometry) and automorphic-type data (global sections of the topos) can be posed internally. The Langlands program classifies what cannot be globally trivialized and how those failures organize into rigid spectral data. The tower topos instantiates this architecture: local symmetries exist at every level, very few globalize, and the pattern of globalization failure is classified by the Čech cohomology of Section 10.

This alignment is structural, not a claim to solve Langlands-type problems. The tower provides a concrete instantiation of the categorical environment in which such problems are naturally posed—an environment usually constructed abstractly. The natural reconnection point between the RH branch of this series (Paper II) and the tower branch (Papers III–V) lies precisely here: the Riemann zeta function should appear as the L-function of the simplest global section of $\mathcal{T}$ —the one associated with the $\mathbb{Z}_2$ holonomy itself—with zeros on the critical line as a theorem of the topos rather than a standalone analytic result. The physical identification of the globalizable groups with gauge groups of the Standard Model is deferred to Paper VI.

12 The Tower as Fixed Point

The topos structure established in Sections 9 and 10 enables a precise statement of the tower’s self-referential character, completing the arc begun in Paper III’s self-grounding theorem.

Theorem 12.1 (Fixed-point character). *The tower topos $\mathcal{T}$ is a fixed point of its own generative principle in the following sense:*

- (i) *The Möbius connection generates $\mathbb{Z}_2$, which generates the topos $\mathbb{Z}_2\text{-Set}$ (Paper I $\rightarrow$ Paper III).*
- (ii) *Within $\mathbb{Z}_2\text{-Set}$, Lawvere’s fixed-point theorem [8] reproduces the $\mathbb{Z}_2$ involution that determines the connection (Paper III, self-grounding).*
- (iii) *The connection generates the Cayley–Dickson tower, whose constraint geometry forms the topos $\mathcal{T}$ (Sections 9–10).*
- (iv) *The internal logic of $\mathcal{T}$ is the three-valued classifier $\Omega = \{\text{true}, \text{false}, \text{swap}\}$ —the same classifier as $\mathbb{Z}_2\text{-Set}$.*
- (v) *The obstruction structure of $\mathcal{T}$ (swap values, non-globalizable sections, Čech cohomology classes) is the categorical expression of the representational obstructions that generated $\mathcal{T}$ in the first place.*
- (vi) *The functional architecture (addressing tiers, Fano organization, selectivity stabilization) recapitulates at each level the structure of the tower as a whole: levels are indexed like integers, tiers grow linearly while content grows exponentially, inheritance across tiers mirrors inheritance across levels, and the stable octonionic core with widening periphery mirrors the NDA/post-NDA regime structure.*

The tower does not merely contain self-referential structure. It *is* the fixed point of iterated self-reference: the object whose internal logic, when applied to its own obstruction structure, reproduces the object.

12.1 Relationship to Paper III

Paper III established self-grounding at the base level: $\omega \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_2\text{-Set} \rightarrow \text{Lawvere} \rightarrow \mathbb{Z}_2 \rightarrow \omega$. Theorem 12.1 establishes self-grounding at the *tower* level: the entire constraint hierarchy, when described categorically, reproduces the logic that generated it. The base-level fixed point nests inside the tower-level fixed point as its innermost layer—exactly as $\mathbb{Z}_2\text{-Set}$ nests inside $\mathcal{T}$ as a sub-topos.

12.2 Fiber duality as fixed-point signature

The discrete/continuous duality of Section 8 is the fiber-level manifestation of the fixed-point character. The tower viewed discretely (level by level) and continuously (as a completed limit object) yield the same structural content, reconciled by $\mathbb{R}$ as the scalar fixed point. This reconciliation is not imposed—it is the inevitable consequence of a self-referential structure whose internal logic is stable under its own iteration.

12.3 Scale invariance resolved

Non-termination of the tower (Paper IV, Theorem 12.1(iii)) reflects stabilization of generative form, not incompleteness. The fixed-point character ensures that no additional

type of structure can appear at higher levels that is not already present as a pattern at lower levels—though the specific instantiation at each level is new. The distinction between “no new types” and “no new content” is the distinction between the form being fixed and the content being infinite.

The emergence of physical scale from this self-similar regime—the mechanism by which intrinsic comparison structures and reference scales arise—is the subject of subsequent work.

13 Discussion

13.1 What this paper establishes

Remark 13.1 (On the scope of potential objections). The results of this paper divide into two categories with different evidential status. The *structural findings*—the component decomposition, functional tiers, Fano organization, kernel conservation, spectral rigidity, and transparency—are verified exhaustively at each computed level. These are mathematical facts about specific finite-dimensional algebras, not conjectures. No objection to the extensibility of counting formulas to higher levels affects the validity of these findings at the levels where they are established.

The *structural claims*—that counting formulas extend to all levels, that the topos axioms hold in full generality, that group completion yields exactly the NDA chain—are supported by six levels of verification and by structural arguments, but are not proved in the sense of a complete inductive proof. The distinction matters: the constraint geometry of the post-NDA Cayley–Dickson tower is a new mathematical object whose systematic study begins here. The exploration of this territory is the primary contribution; the conjectured laws governing it are offered as the best current description of what the exploration has found.

1. The post-NDA constraint geometry organizes into a hierarchical functional architecture with exact formulas, topological protection, Fano-plane organization, and logarithmic compression (Sections 1–7).
2. The fiber simultaneously supports discrete and continuous views, reconciled by $\mathbb{R}$ as scalar fixed point (Section 8).
3. This architecture satisfies the axioms of an elementary topos, verified explicitly against the computationally generated structure (Section 9).
4. The natural site structure of this topos is determined by kernel coherence, with covering families given by connected components and obstruction tracked by swap-valued Čech cocycles (Section 10).
5. Group completion of the topos’s automorphism structure extracts symmetry groups as mathematical necessities (Section 11).
6. The tower is a fixed point of its own generative principle, with the internal logic of $\mathcal{T}$ reproducing the obstruction structure that generated it (Section 12).

13.2 Falsifiability at the topos level

Physical properties relative to this framework fall into three structurally distinct classes:

- (A) **Fully globalizable properties:** Conservation laws, fundamental symmetry groups, spectral constraints. These correspond to global sections of $\mathcal{T}$. Empirical violation of any of these would *falsify* the framework.

- (B) **Consistently local (non-globalizable) properties:** Chirality, contextuality, phase selection, sector-dependent behavior. These correspond to local sections that cannot glue globally, classified by cohomology. They are *predicted* by the obstruction structure; their presence *supports* the framework.
- (C) **Inconsistent local properties:** Violations of internal logic, kernel/degree relations, or pairing structure. These would be incompatible with $\mathcal{T}$ even locally and would *falsify* the framework.

The framework is falsified by inconsistency, not by locality. A world with no non-globalizable observables would be more suspicious under this framework than one with structured obstruction.

13.3 Forward directions

Open Question 13.2 (L-functions and spectral theory). Global sections of $\mathcal{T}$ have associated obstruction spectra encoding globalization failure across levels. These should be L-functions in a precise sense. The L-function of the simplest global section (associated with the $\mathbb{Z}_2$ holonomy) should reproduce the Riemann zeta function, reconnecting Paper II’s result to the tower’s internal logic. This is the subject of Paper VI.

Open Question 13.3 (Physical identification). The globalizable symmetry groups $\mathbb{Z}_2, \mathrm{SU}(2), G_2 \supset \mathrm{SU}(3)$ invite identification with the gauge groups of the Standard Model: $U(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$. The chirality decomposition of the constraint geometry invites comparison with the chiral structure of the weak interaction. Whether these identifications are exact—and what physical predictions follow—is the subject of Paper VI.

Open Question 13.4 (Metric emergence). Metrics require global, obstruction-free comparability, while the tower encodes controlled failures of exactly that. Metric emergence is conditional on obstruction stabilization, identity sharpening within a chosen sector, and a distinguished reference section. This is genuinely hard, honestly deferred, and the hardness is itself evidence that the framework is doing something structurally real—metrics should be the *last* thing to emerge, after states, symmetries, and spectra.

Open Question 13.5 (Prime structure from the tower). The prime numbers appear in this framework as irreducible dimensional obstructions: the algebraic irrational defined by $x^p = x + 1$ for prime p is the scalar trace of an intrinsically p -dimensional relational process that cannot be decomposed into lower-dimensional pieces. The primes’ role as generators in the RH paper (Paper II) should be the same role they play as irreducible obstructions in the Cayley–Dickson tower. If the tower’s internal logic is genuinely self-referential, the prime factorization structure—not just the gauge groups—should emerge from it.

Data and code availability

The computational pipeline used to generate and verify the results reported in this paper—including the Cayley–Dickson algebra implementation, zero-divisor search, component analysis, structural prediction engine, and Fano-enriched tower generator—is available at:

<https://github.com/TheAlphasapien/Mobius-homotopy-tower>

The repository includes verified structural data for all levels through dimension 512 (level 6), comprising multiplication tables, zero-divisor pair catalogs, component decompositions with full metadata, and the prediction engine that reproduces all reported counting formulas. The prediction engine generates the complete structural census at arbitrary tower depth without floating-point computation; the brute-force data serves as ground truth for validation.

References

- [1] A. K. Morgan. The Möbius connection as canonical generator of arithmetic structure. Working paper, 2026.
- [2] A. K. Morgan. Orientation, fixed-point loci, and the Riemann Hypothesis. Working paper, 2026.
- [3] A. K. Morgan. Non-orientable foundations: geometric resolution of set-theoretic paradoxes via the Möbius connection. Working paper, 2026.
- [4] A. K. Morgan. Tower generation from Möbius holonomy: canonical resolution of representational obstruction. Working paper, 2026.
- [5] G. Moreno. The zero divisors of the Cayley–Dickson algebras over the real numbers. *Bol. Soc. Mat. Mexicana*, 4(3):13–28, 1998.
- [6] R. D. Schafer. *An Introduction to Nonassociative Algebras*. Academic Press, 1966.
- [7] S. Mac Lane and I. Moerdijk. *Sheaves in Geometry and Logic*. Springer, 1992.
- [8] F. W. Lawvere. Diagonal arguments and cartesian closed categories. *Lecture Notes in Math.*, 92:134–145, 1969.